

# *The Ultimate Survival Guide to Grouped Data: The Big 3 M's (A Masterclass in Statistics)*



Swipe to unlock your ultimate math toolkit! ➡



## **MEAN** *(The Average)*

The sum of all values divided by the total number. The most frequently used measure of central tendency.



## **MODE** *(The Popular One)*

The value that occurs most often. The observation with the maximum frequency.



## **MEDIAN** *(The Middle Ground)*

The value of the middle-most observation when data is arranged in order.

Today, we level up from ungrouped to **GROUPED** data! 🚀

# Deep Dive: The Mean ( $\bar{x}$ ) & The Class Mark

## The Problem

| 1              | 2                   |
|----------------|---------------------|
| Class Interval | Frequency ( $f_i$ ) |
| 10-25          | 2                   |
| 25-40          | 3                   |

How do we multiply an interval by a frequency?!

## The Solution

We need a single representative number for the whole class!

Enter the Class Mark ( $x_i$ ).

$$\text{Class Mark} = \frac{\text{Upper limit} + \text{Lower limit}}{2}$$

$$\text{For } 10-25, x_i = \frac{10 + 25}{2} = 17.5$$

# The 3 Mean Methods: Choose Your Weapon! ✂



## Direct Method

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

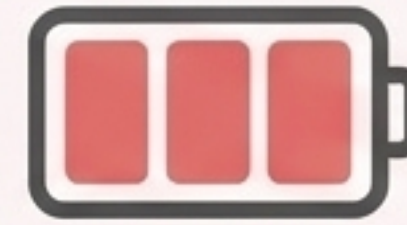
Best when  $x_i$  and  $f_i$  are small numbers.



## Assumed Mean

$$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$$

When numbers are getting large. We subtract an assumed mean 'a' to get deviations ( $d_i = x_i - a$ ).

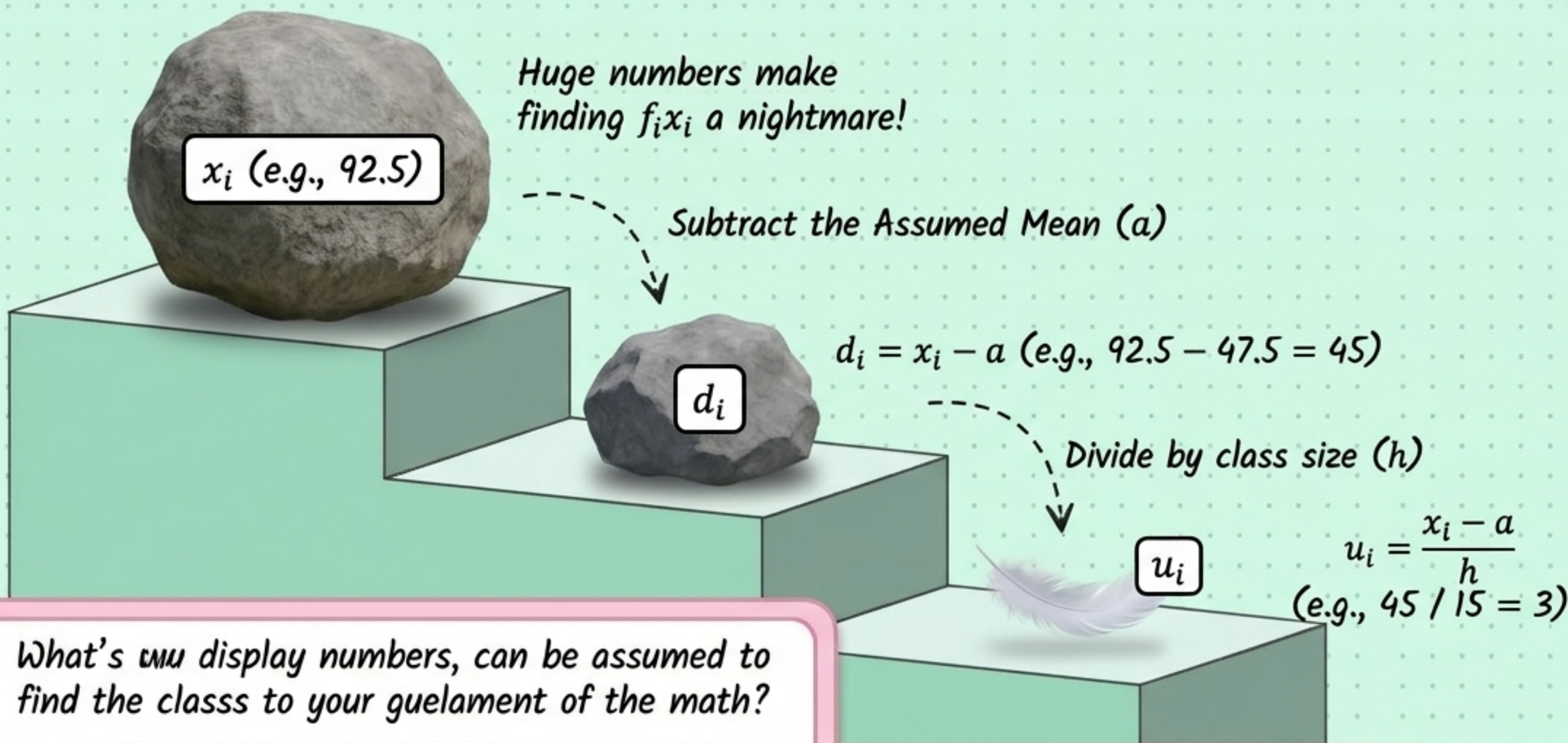


## Step-Deviation

$$\bar{x} = a + h \left( \frac{\sum f_i u_i}{\sum f_i} \right)$$

When  $d_i$  values have a common factor 'h'.

# Why step-deviation? Making heavy math light!



By scaling down the numbers, the multiplication ( $f_i u_i$ ) becomes mental math!  
At the end, the formula just scales it back up:  $\bar{x} = a + hu$ .

# Spotlight: The Mode & The Modal Class 👑

|       |         |
|-------|---------|
| 10-25 | Freq: 2 |
| 25-40 | Freq: 3 |
| 40-55 | Freq: 7 |
| 55-70 | Freq: 6 |

**Step 1: Spot the biggest frequency!**

The class interval with the highest frequency is your Modal Class. Everything in the formula revolves around this single row!

# Anatomy of the Mode Formula

$f_0$ : Frequency of the class preceding (before) the modal class.

$f_1$ : Frequency of the modal class (The Star!).

$f_2$ : Frequency of the class succeeding (after) the modal class.

$$\text{Mode} = l + \left[ \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

$l$ : Lower limit of the modal class.

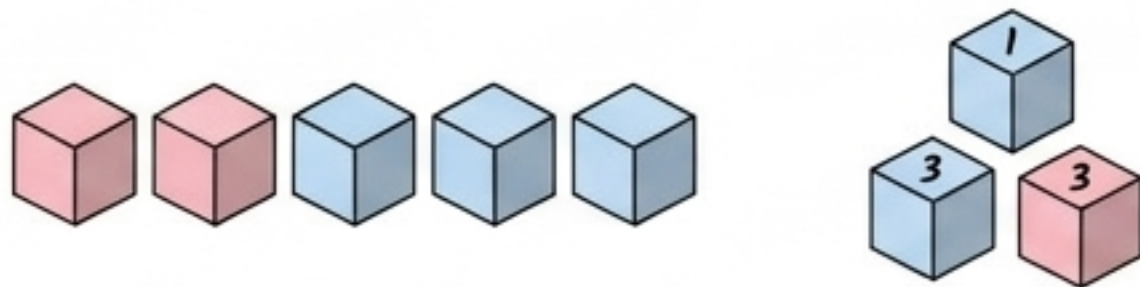
$h$ : Class size.

Remember:  $f_0, f_1, f_2$  are just in numerical order!  
Before (0), Current (1), After (2).

# The Middle Ground: Finding the Median

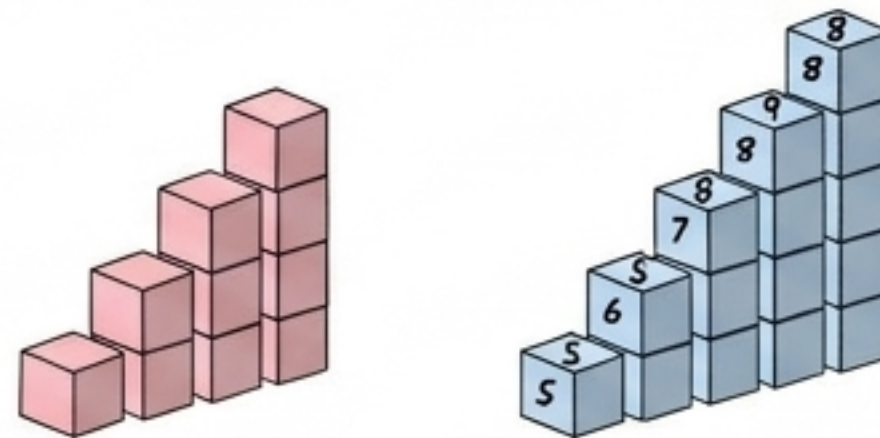
To find the exact middle of grouped data, we can't just look at normal frequencies. We have to see how the data piles up.

## Normal Frequency (f)



Tells us how many are in one specific class.

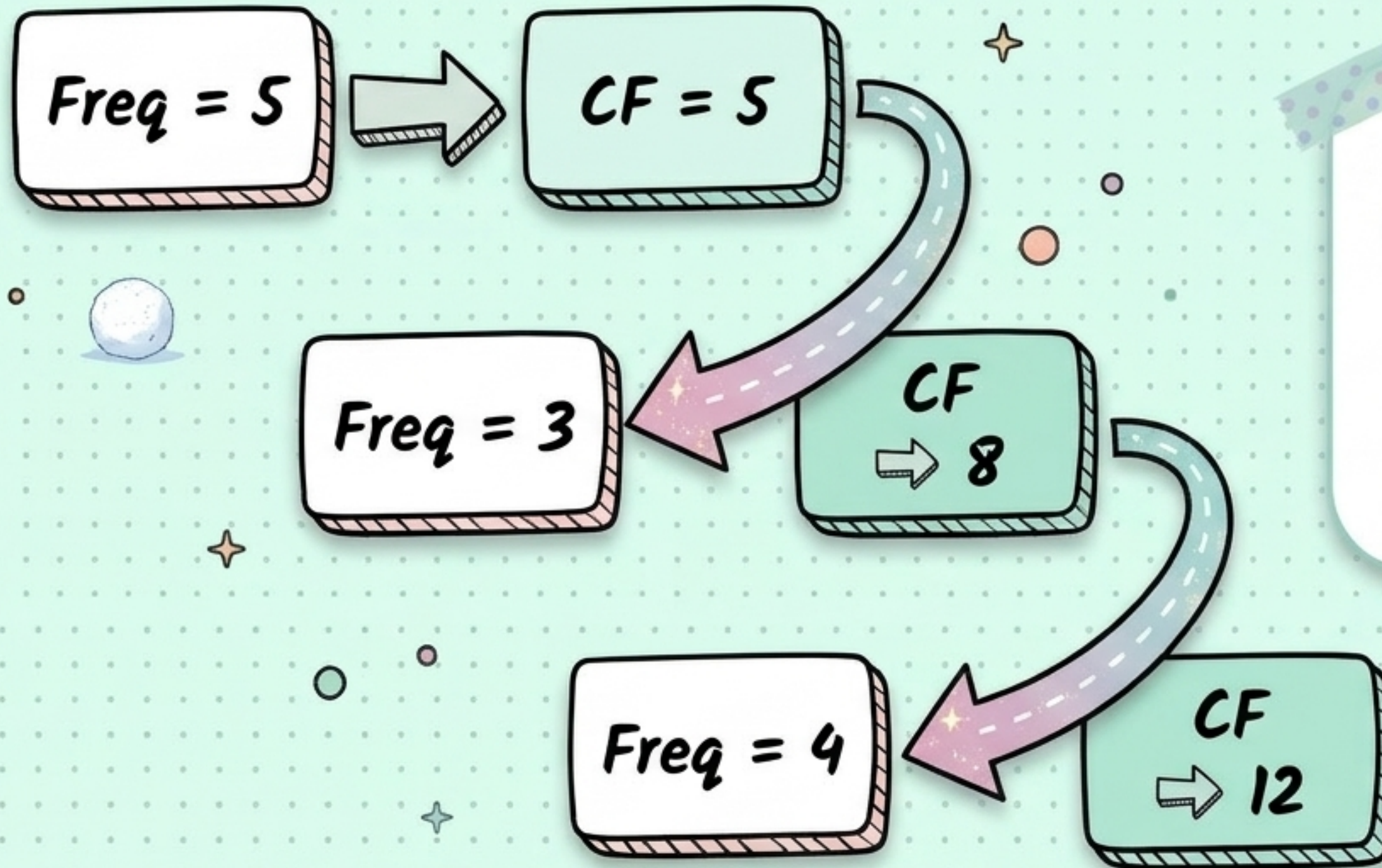
## Cumulative Frequency (cf)



Tells us how many students scored less than the upper limit.

The Median Class is the class whose Cumulative Frequency (cf) is greater than and nearest to  $N/2$  (where  $N$  is total frequency).

# The CF Accumulator: Snowballing Frequencies 🌨️



Just keep adding diagonally! Like a snowball rolling down a hill, it gathers all the frequencies that came before it.

# Anatomy of the Median Formula

$\ell$ : Lower limit  
of median  
class.

$n$ : Number of  
observations  
(Total sum of  
frequencies).

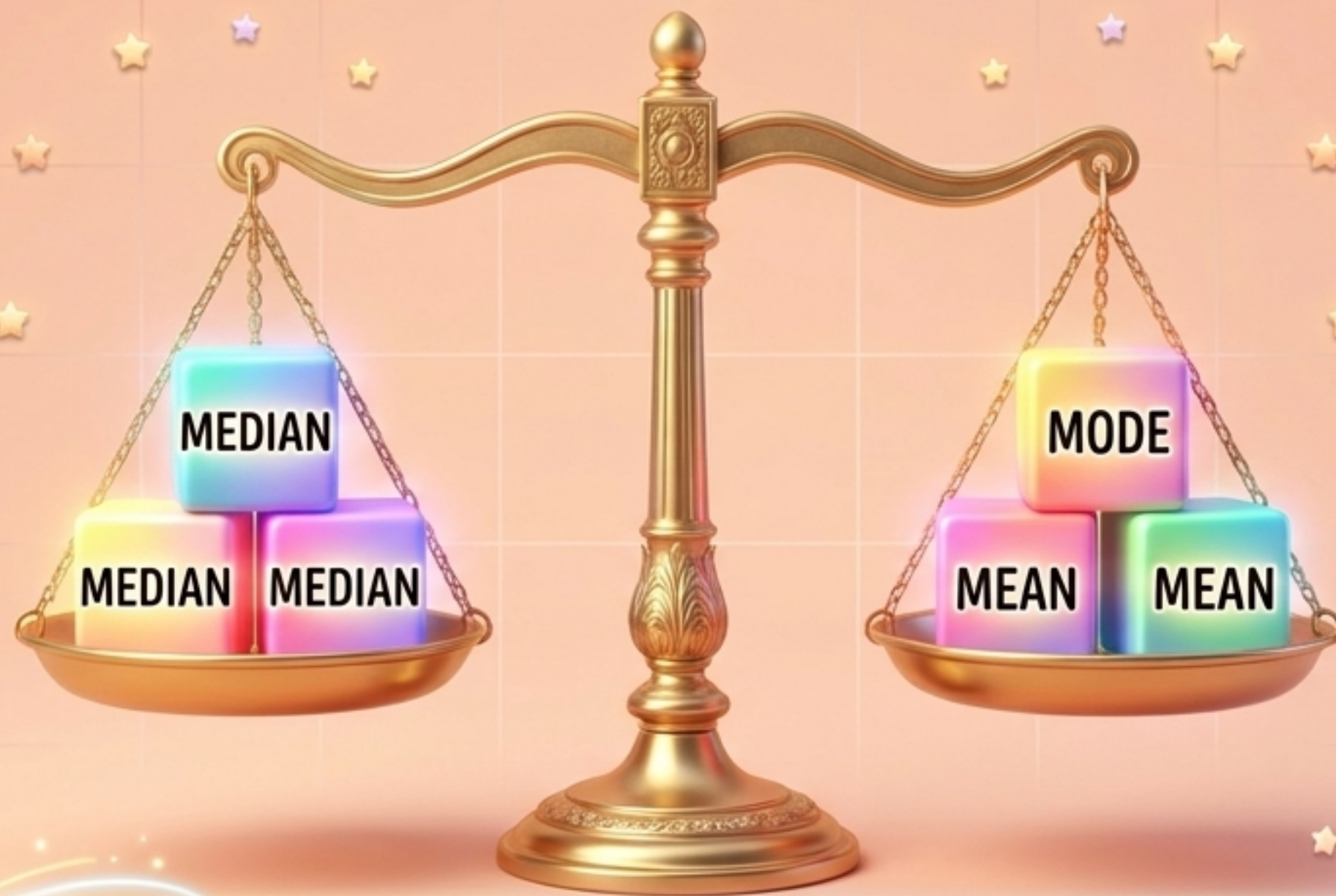
$$\text{Median} = \ell + \left[ \frac{\frac{n}{2} - cf}{f} \right] \times h$$

$cf$ : Cumulative  
frequency of the class  
**PRECEDING** (before)  
the median class!

$f$ : Frequency  
of the median  
class itself.

$h$ : Class size.

# The Boss Level: The Empirical Relationship



$$3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$$

 Got 2 out of 3?  
Use this cheat code  
to instantly find the  
missing measure  
without doing the  
heavy math!

# The Ultimate Cheat Sheet: Which 'M' Do I Use?

## The MEAN

Comparing overall performance (e.g., average exam scores of two different schools).

! Gets easily messed up by extreme outliers!

## The MEDIAN

Finding the "typical" scenario ignoring crazy extremes (e.g., typical productivity rate of workers, typical country wage).

✓ Safest bet when data is skewed.

## The MODE

Finding the most popular item (e.g., most watched TV show, most common shoe size in a store).

👑 The king of consumer demand.

Save this post to ace your stats exam! 📌💖